

# DESHAN SIRIWARDHANA

## Full-Stack Developer

BSc (Hons) Software Engineering | CINEC Campus

### CONTACT

#### PHONE

(+94) 76 977 6315

#### EMAIL

[deshandhakshitha16@gmail.com](mailto:deshandhakshitha16@gmail.com)

#### LOCATION

Kegalle / Colombo, Sri Lanka

#### PORTFOLIO

[deshansiriwardhana.me](https://deshansiriwardhana.me)

#### GITHUB

[github.com/deshan2004](https://github.com/deshan2004)

#### LINKEDIN

<http://www.linkedin.com/in/deshan-siriwardhana>

### EDUCATION

#### CINEC Campus

2024 - Present | Malabe, Sri Lanka

BSc (Hons) in Software Engineering

**Key coursework:** Data Structures & Algorithms, Software Construction and Evolution, Software Quality Assurance, Database Management Systems, Mobile Computing, Advanced Web Technologies, Software Testing.

#### Swarna Jayanthi College

2022 | Kegalle, Sri Lanka

G.C.E. Advanced Level - Physical Science Stream

### TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript (ES6+), Java, Python, C#, PHP, C, Kotlin, Dart

**Frontend & Mobile:** Next.js, React.js, Angular, Vue.js, Flutter, Tailwind CSS, Bootstrap, HTML5, CSS3

**Backend & APIs:** Node.js, Express.js, RESTful APIs, WebRTC, ASP.NET Core, PHP, Nginx, Python, Java

**Databases & Cloud:** PostgreSQL, Supabase, MySQL, MSSQL, MongoDB, Firebase, Oracle

**Testing & QA:** Unit Testing, JUnit 5, Apache JMeter, Selenium WebDriver, Requirements Traceability (RTM)

**DevOps & Tools:** Git, GitHub, VS Code, Postman, IntelliJ IDEA, Android Studio, Figma, Vercel

### OTHER SKILLS

**Leadership & Management:** Team Leadership, Agile / Scrum Practices, Project Planning, Cross-Functional Collaboration

**Problem Solving:** Analytical Thinking, Debugging, Critical Decision Making, Requirements Analysis

**Communication & Interpersonal:** Technical Documentation, Stakeholder Communication, Public Speaking, Client Engagement

**Workplace Competencies:** Time Management, Adaptability, Active Listening, Conflict Resolution

### PROFILE

Versatile Full-Stack Software Engineer and Startup Co-Founder with a solid foundation bridging core web fundamentals (HTML5, CSS3, modern JavaScript) to advanced enterprise architectures (Next.js, React.js, TypeScript, Node.js, .NET Core). Proven track record of turning requirements into production-ready web and mobile platforms integrating real-time WebSockets, AI/OCR services, and scalable cloud databases (PostgreSQL, Supabase, Firebase). Dedicated to building accessible, clean, and type-safe systems backed by automated QA testing and agile delivery.

### EXPERIENCE

#### Co-Founder & Technical Lead

2026 - Present

CodeCraft Solutions

Co-founded a software solutions venture, directing technical strategy, system architecture, and modern UI/UX workflows for bespoke web and mobile applications.

Architected and shipped responsive full-stack solutions for local business clients, translating stakeholder requirements into scalable, maintainable software products.

### FEATURED PROJECTS

#### ShelfLife AI - AI-Powered Inventory & Expiry Loss Prevention Platform

React.js(vite) · Node.js · Firebase · Tesseract.js · REST API · PWA · JavaScript

- Engineered an AI-OCR scanning module using Tesseract.js/Vision AI to auto-extract expiration dates directly from packaging, cutting manual stock entry time by 90%.
- Developed a real-time FEFO (First-Expired, First-Out) inventory engine with tiered alerts (30/7/2-day windows) and batch-level supplier tracking to eliminate expiry losses.
- Architected automated Flash Sale triggers that dynamically apply promotional discounts (BOGO, 30-50% off) on near-expiry items to recover at-risk revenue.
- Integrated an AI conversational assistant & recipe engine to recommend bundled promotions and waste-reduction strategies for food/retail inventory.
- Built an Offline-First PWA architecture with background data sync and role-based access control (User & Admin portals) for uninterrupted shop-floor operations.

#### Kingswood Connect - Real-Time Student Attendance & Management Platform

React.js(Vite) · Node.js · Express.js · Firebase · Socket.IO · Tailwind CSS · JavaScript

- Engineered a hardware-free, full-stack student attendance and administrative management platform for educational institutions.
- Architected a real-time mobile scanning integration using Socket.IO WebSockets, allowing smartphone cameras to function as live ID/QR scanners for desktop dashboards.
- Configured a Node.js and Express backend integrated with Firebase (Firestore & Auth) to enforce multi-role authorization (Admin, Teacher, Student) and secure data management.
- Designed an administrative web portal for live attendance analytics, automated student fee tracking, class scheduling, and a soft-delete data recovery pipeline using React and Tailwind CSS.

## CERTIFICATIONS

### Basics of Machine Learning Algorithms (Level 7)

Athena Global Education

*Decision Trees, Naive Bayes, Predictive Modeling*

### Operate with DevOps

Microsoft

*DevOps Practices, Continuous Integration & Delivery, Agile Workflows*

### Explore Microsoft 365 Security Foundations

Microsoft

*Identity & Access Management, Threat Protection, Cloud Security*

### CIRS 2026 Workshop on Authentic Learning

CINEC Campus

*Real-World Experiences That Build 21st-Century Skills*

## LANGUAGES

**English:** Professional Working Proficiency

**Sinhala:** Native / Bilingual Proficiency

## REFERENCES

Mr. D.P.N. Hasaranga

Lecturer

Faculty of Computing, CINEC Campus

Email: [naveenhasaranga.0989@gmail.com](mailto:naveenhasaranga.0989@gmail.com)

Phone: (+94) 76 285 7503

Mr. Hasira Koswatta

Lecturer

Faculty of Computing, CINEC Campus

Email: [hasirakoswatta09@gmail.com](mailto:hasirakoswatta09@gmail.com)

Phone: (+94) 70 108 2872

### Neth-Sawan - AI-Powered Assistive Web Platform

*React.js(vite) · JavaScript · Node.js · WebRTC · Google Generative AI SDK · Firebase*

- Engineered a real-time assistive communication portal tailored for hearing-impaired users to bridge daily communication barriers.
- Implemented peer-to-peer real-time video communication using WebRTC paired with Firebase signaling servers.
- Integrated Google Generative AI SDK to process camera feeds into instant visual contextual descriptions, ambient event detections, and scannable visual alerts.
- Developed a high-contrast, accessible UI in React.js and TypeScript following WCAG accessibility guidelines.

### KethPiyasa - Direct Farmer-to-Buyer B2B Agricultural Marketplace

*Next.js · TypeScript · Node.js · Express.js · Firebase · Tailwind CSS*

- Engineered a full-stack B2B digital marketplace designed to streamline Sri Lanka's agricultural supply chain by bridging smallholder farmers directly with commercial buyers.
- Implemented role-based portals for Farmers, Buyers, Logistics, and Admins featuring real-time crop bidding, counter-offer negotiations, and interactive geographic discovery maps.
- Integrated an escrow holding payment workflow paired with automated invoice generation and QR code delivery verification to secure transactions.
- Architected a trilingual localization system (English, Sinhala, Tamil) and live economic center baseline price tracking to ensure market transparency and accessibility.

### RIGA Academy - Dance Academy Management & Online Enrollment Platform

*Next.js · TypeScript · Firebase (Auth & Firestore) · Tailwind CSS · Framer Motion*

- Engineered a full-stack dance academy platform to streamline student enrollments, class scheduling, event showcases, and studio space management.
- Integrated Firebase Authentication and Firestore DB to enforce multi-role access control (Admin & Student), securing user data and administrative permissions.
- Developed an online enrollment & verification workflow, allowing students to register for dance packages, upload payment slips, and track registration status via a dedicated dashboard.
- Architected an intuitive Admin Portal for live attendance tracking, class scheduling, payment slip approvals, user role management, and inquiry handling.
- Crafted a modern, responsive UI/UX using Next.js, TypeScript, Framer Motion, and Tailwind CSS with dynamic animations, interactive schedules, and media galleries.

## LEADERSHIP & VOLUNTEERING

### Active Member

2026 - Present

*AIESEC in CINEC*

Contributing as an active member supporting youth leadership initiatives and chapter activities.